Ex:No: 15

DCL COMMANADS– GRANT, REVOKE

1)

```
mysql> CREATE TABLE Student (
    ->     ID INT PRIMARY KEY,
    ->     Name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.841 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Sandra');
Query OK, 1 row affected (0.120 sec)

mysql>
mysql> CREATE USER 'student_user'@'localhost'
    -> IDENTIFIED BY '1234';
Query OK, 0 rows affected (0.569 sec)

mysql> GRANT SELECT ON experiment15.Student
    -> TO 'student_user'@'localhost';
Query OK, 0 rows affected (0.095 sec)

mysql> SHOW GRANTS FOR 'student_user'@'localhost';
+------------------------------------------------------------------------+
| Grants for student_user@localhost                                      |
+------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `student_user`@`localhost`                       |
| GRANT SELECT ON `experiment15`.`student` TO `student_user`@`localhost` |
+------------------------------------------------------------------------+
2 rows in set (0.032 sec)
```

2)

```
mysql> GRANT INSERT ON experiment15.Student
    -> TO 'student_user'@'localhost';
Query OK, 0 rows affected (0.053 sec)

mysql> REVOKE INSERT ON experiment15.Student
    -> FROM 'student_user'@'localhost';
Query OK, 0 rows affected (0.131 sec)

mysql> SHOW GRANTS FOR 'student_user'@'localhost';
+------------------------------------------------------------------------+
| Grants for student_user@localhost                                      |
+------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `student_user`@`localhost`                       |
| GRANT SELECT ON `experiment15`.`student` TO `student_user`@`localhost` |
+------------------------------------------------------------------------+
2 rows in set (0.005 sec)
```

Ex:No: 14

TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

1)

```
mysql>
mysql> CREATE TABLE Class (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.980 sec)

mysql>
mysql> INSERT INTO Class VALUES
    -> (1,'alpha'),
    -> (2,'beta'),
    -> (3,'gamma'),
    -> (4,'delta'),
    -> (5,'echo');
Query OK, 5 rows affected (0.161 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> UPDATE Class SET name='bravo' WHERE id=5;
Query OK, 1 row affected (0.061 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.007 sec)

mysql>
mysql> INSERT INTO Class VALUES(6,'uppal');
Query OK, 1 row affected (0.004 sec)

mysql> SAVEPOINT B;
Query OK, 0 rows affected (0.003 sec)
```

2)

```
mysql>
mysql> INSERT INTO Class VALUES(7,'balu');
Query OK, 1 row affected (0.004 sec)

mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> SELECT * FROM Class;
+----+-------+
| id | name  |
+----+-------+
|  1 | alpha |
|  2 | beta  |
|  3 | gamma |
|  4 | delta |
|  5 | bravo |
|  6 | uppal |
|  7 | balu  |
+----+-------+
7 rows in set (0.015 sec)

mysql>
mysql> ROLLBACK TO B;
Query OK, 0 rows affected (0.011 sec)

mysql>
mysql> SELECT * FROM Class;
+----+-------+
| id | name  |
+----+-------+
|  1 | alpha |
|  2 | beta  |
|  3 | gamma |
|  4 | delta |
|  5 | bravo |
|  6 | uppal |
+----+-------+
6 rows in set (0.005 sec)
```

Ex:No: 13

Simple programming usingCASE and LOOP

1)

```
MySQL 9.6 Command Line Cli

    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE income VARCHAR(20);
    ->
    ->     CASE
    ->         WHEN monthly_value <= 4000 THEN
    ->             SET income = 'Low Income';
    ->         WHEN monthly_value <= 5000 THEN
    ->             SET income = 'Avg Income';
    ->         ELSE
    ->             SET income = 'High Income';
    ->     END CASE;
    ->
    ->     RETURN income;
    -> END //
ERROR 1304 (42000): FUNCTION income_level already exists
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT income_level(3500) AS Income_Level;
+--------------+
| Income_Level |
+--------------+
| Low Income   |
+--------------+
1 row in set (0.003 sec)

mysql> SELECT income_level(4500) AS Income_Level;
+--------------+
| Income_Level |
+--------------+
| Avg Income   |
+--------------+
1 row in set (0.007 sec)

mysql> SELECT income_level(6000) AS Income_Level;
+--------------+
| Income_Level |
+--------------+
| High Income  |
+--------------+
```

2)

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION check_income()
    -> RETURNS VARCHAR(30)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE income INT DEFAULT 1000;
    ->     DECLARE result VARCHAR(30) DEFAULT '';
    ->
    ->     myloop: LOOP
    ->         SET income = income + 1000;
    ->
    ->         IF income < 4000 THEN
    ->             ITERATE myloop;
    ->         END IF;
    ->
    ->         SET result = 'Income reached 4000';
    ->         LEAVE myloop;
    ->     END LOOP;
    ->
    ->     RETURN result;
    -> END //
Query OK, 0 rows affected (0.109 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT check_income() AS Result;
+---------------------+
| Result              |
+---------------------+
| Income reached 4000 |
+---------------------+
```

Ex:No: 12

Simple Programming using REPEAT, WHILE

1)

```
mysql> USE DBMS;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION build_string()
    -> RETURNS VARCHAR(100)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE x INT DEFAULT 1;
    ->     DECLARE result VARCHAR(100) DEFAULT '';
    ->
    ->     WHILE x <= 5 DO
    ->         SET result = CONCAT(result, x);
    ->         SET x = x + 1;
    ->     END WHILE;
    ->
    ->     RETURN result;
    -> END //
Query OK, 0 rows affected (1.438 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT build_string();
+----------------+
| build_string() |
+----------------+
| 12345          |
+----------------+
1 row in set (0.135 sec)
```

2)

```
mysql> CREATE FUNCTION calculate_income()
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE income INT DEFAULT 1000;
    ->
    ->     REPEAT
    ->         SET income = income + 1000;
    ->     UNTIL income >= 4000
    ->     END REPEAT;
    ->
    ->     RETURN income;
    -> END //
Query OK, 0 rows affected (0.186 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT calculate_income();
+--------------------+
| calculate_income() |
+--------------------+
|               4000 |
+--------------------+
1 row in set (0.233 sec)
```

Ex. No.: 11

DATABASE WITH AUTO_INCREMENT SEQUENCES

1)

```
mysql> SELECT * FROM DBMS_Stud;
+-------------+-------+------------+
| Register_No | Name  | Department |
+-------------+-------+------------+
|           1 | Arun  | CSE        |
|           2 | Bala  | ECE        |
|           3 | Divya | CSE        |
|           4 | Meena | IT         |
+-------------+-------+------------+
4 rows in set (0.098 sec)
```

2)

```
mysql> SELECT * FROM DBMS_Stud;
+-------------+-------+------------+
| Register_No | Name  | Department |
+-------------+-------+------------+
|           1 | Arun  | CSE        |
|           2 | Bala  | ECE        |
|           3 | Divya | CSE        |
|           4 | Meena | IT         |
|          10 | Kiran | IT         |
+-------------+-------+------------+
5 rows in set (0.008 sec)
```

3)

```
DBMS_Stud | CREATE TABLE `dbms_stud` (
`Register_No` int NOT NULL AUTO_INCREMENT,
`Name` varchar(30) DEFAULT NULL,
`Department` varchar(20) DEFAULT NULL,
PRIMARY KEY (`Register_No`)
ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

4)

```
1 row in set (0.090 sec)

mysql> ALTER TABLE DBMS_Stud
    -> MODIFY Register_No INT PRIMARY KEY;
ERROR 1068 (42000): Multiple primary key defined
mysql> SHOW CREATE TABLE DBMS_Stud;
```

5)

```
| DBMS_Stud | CREATE TABLE `dbms_stud` (
  `Register_No` int NOT NULL AUTO_INCREMENT,
  `Name` varchar(30) DEFAULT NULL,
  `Department` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`Register_No`)
) ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

6)

```
mysql> SELECT * FROM DBMS_Stud;
+-------------+-------+------------+
| Register_No | Name  | Department |
+-------------+-------+------------+
|           1 | Arun  | CSE        |
|           2 | Bala  | ECE        |
|           3 | Divya | CSE        |
|           4 | Meena | IT         |
|          10 | Kiran | IT         |
|          11 | Ravi  | CSE        |
+-------------+-------+------------+
6 rows in set (0.162 sec)
```